

2. solve $x \log x - 1.2 = 0$ by Regula falsi or false position method. Ans: 2.7405

3. solve $x - \cos x = 0$ by false position method.

Ans: 0.739079

1. Define the system of equation, write condition for convergence of system of equation in Gauss Seidel method.

$$\therefore \begin{aligned} x_1 &= 10 \\ x_2 &= 10 \\ x_3 &= 10 \end{aligned}$$

HW. Solve the following system of equation: $28x + 4y - z = 32$,
 $x + 3y + 10z = 24$, $2x + 17y + 4z = 35$ using Gauss-Siedel
method. (Ans: $x = 0.994$, $y = 1.507$, $z = 1.849$).

How

2. Prove that

$$(i). E^{-1/2} = \mu - 1/2 \delta.$$

3. Prove that $\Delta = 1/2 \delta^2 + \delta \sqrt{1 + \frac{\delta^2}{4}}$

Find $y(20)$ from the following table. using divided difference.

<u>x</u>	<u>y</u>	<u>ϕ</u>
12	146	115
18	836	
22	19481	
24	2796	
32	9236	

H.W.
3.

The population of a certain town is given below. Find the
rate of growth of population in 1931, 1941, & 1971.

derivative.

<u>year</u>	<u>Population in (1000)</u>
1931	40.62
1941	60.8
1951	79.95
1961	103.56
1971	132.65

4. Find the derivative of the function at $x=900$ from the following table.

x	y
0	135
300	149
600	157
900	183
1200	201
1500	205
1800	193

H.W

1. Prob

$$\int_0^6 \frac{dx}{1+x}$$

using trapezoidal, simpsons $\frac{1}{3}$ rd, simpsons $\frac{3}{8}$ th

let $h=1$.

How
1. solve $\frac{dy}{dx} = x^2 - y$, $y(0) = 1$ at $x = 0.1$ & $x = 0.2$ using
euler's method.

2. Solve $\frac{dy}{dx} = x^2 + y^2$, $y(0) = 1$ at $x = 0.1$ & $x = 0.2$ using modified euler's method.

3. Solve $\frac{dy}{dx} = x^2 y - 1$, $y(0) = 1$ at $x = 0.1$ using Taylor series method.